

FARGO HECTOR, ND, USA

WMO: 727530

Lat: 46.925N

Lon: 96.811W

Elev: 900

StdP: 14.22

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14914

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.7	-13.9	-25.2	1.4	-17.7	-20.5	1.8	-13.4	33.9	5.7	29.7	10.7	7.5	310	0.681

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.9	90.0	72.4	87.0	70.4	84.3	68.8	75.5	85.4	73.5	83.3	71.5	81.1	14.2	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.4	124.0	82.4	70.0	114.3	80.2	68.0	106.5	77.8	39.7	85.7	37.7	83.3	35.9	81.3	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.3	25.3	22.8	DB	-24.6	95.1	6.3	2.7	-29.1	97.0	-32.8	98.6	-36.3	100.1	-40.9	102.1	(4)
(5)			WB	-24.3	79.3	5.4	1.9	-28.1	80.7	-31.3	81.8	-34.3	82.9	-38.3	84.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.8	9.3	13.5	27.4	43.4	56.7	67.0	71.4	69.1	60.9	46.4	30.3	16.1		
	DBStd	24.24	14.37	13.69	13.46	10.52	9.18	6.85	5.75	5.90	8.67	9.49	11.52	12.84		
	HDD50	5119	1261	1023	706	247	40	0	0	0	12	184	594	1052		
	HDD65	8685	1726	1443	1166	649	284	56	14	27	179	583	1041	1517		
	CDD50	2482	0	0	6	48	249	511	663	591	339	71	4	0		
	CDD65	571	0	0	0	1	28	116	212	153	56	5	0	0		
	CDH74	5670	0	0	1	31	367	1132	2063	1460	560	56	0	0		
	CDH80	1801	0	0	0	6	116	343	694	452	177	13	0	0		
(14) Wind	WSAvg	10.9	11.1	11.4	11.7	12.2	12.0	10.4	9.1	9.1	10.3	11.5	11.3	11.1		
(15) Precipitation	PrecAvg	21.4	0.7	0.6	1.1	1.7	2.8	3.5	2.8	2.5	2.3	1.9	0.8	0.7		
	PrecMax	34.8	1.9	1.7	4.6	5.3	7.9	11.7	7.7	7.5	6.5	7.0	4.6	1.9		
	PrecMin	8.9	0.1	0.1	0.1	0.0	0.5	0.6	0.4	0.2	0.1	0.0	0.0	0.1		
	PrecStd	5.3	0.5	0.4	0.7	1.2	1.7	2.3	1.8	1.5	1.5	1.6	0.9	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.1	46.4	66.4	78.2	89.5	92.6	93.4	92.4	90.7	81.1	63.1	45.3		
		MCWB	36.4	41.1	55.1	57.0	65.8	73.1	75.5	74.4	71.0	62.0	51.9	40.5		
	2%	DB	36.3	40.0	56.9	71.4	82.9	87.6	89.5	88.3	85.1	72.5	56.0	39.0		
		MCWB	33.4	36.4	48.7	53.7	62.7	69.5	73.8	71.5	68.1	58.1	47.3	34.8		
	5%	DB	33.2	36.5	50.3	66.0	77.5	83.9	86.7	85.0	80.6	66.2	50.6	35.5		
		MCWB	31.0	33.8	44.4	51.0	60.2	67.1	71.9	69.9	65.7	55.1	44.1	32.7		
	10%	DB	29.6	33.3	44.5	61.2	73.0	80.8	84.0	82.0	76.0	61.3	45.6	32.6		
		MCWB	27.7	31.4	40.0	48.8	57.6	65.7	70.2	68.3	63.7	52.1	40.0	30.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.2	42.3	57.2	59.6	69.3	76.3	79.2	77.4	74.2	64.6	54.4	40.1		
		MCDB	39.7	45.3	63.0	73.0	82.0	87.2	87.9	88.5	84.7	76.0	61.7	44.6		
	2%	WB	33.6	36.8	49.8	55.1	65.9	73.2	76.5	74.6	70.9	60.2	48.6	35.2		
		MCDB	36.0	39.7	57.2	68.0	78.4	83.9	86.6	84.5	81.5	70.0	55.1	38.2		
	5%	WB	31.0	33.9	44.5	52.3	62.7	70.4	74.2	72.3	68.2	56.2	44.1	32.9		
		MCDB	33.3	36.2	49.8	64.1	73.0	80.4	83.7	81.4	76.8	64.9	50.3	35.3		
	10%	WB	27.7	31.3	39.9	49.6	59.8	67.9	72.2	70.2	65.4	52.6	40.4	30.5		
		MCDB	29.5	33.6	44.5	60.0	69.9	76.8	81.7	79.0	74.5	60.1	45.3	32.6		
(35) Mean Daily Temperature Range	MDBR		16.1	16.9	16.6	20.2	22.1	20.4	20.9	21.8	21.8	18.3	15.8	14.8		
	5% DB	MCDBR	17.2	16.7	23.9	29.7	28.7	24.3	23.6	24.9	26.7	26.4	22.2	17.7		
		MCWBR	15.2	14.0	16.8	16.2	14.6	11.6	11.6	11.7	12.9	15.0	15.3	14.8		
	5% WB	MCDBR	16.8	16.2	23.1	26.4	23.8	20.4	21.1	21.4	22.6	23.6	21.2	16.6		
		MCWBR	14.9	13.7	16.7	15.4	13.5	11.6	12.2	11.9	12.7	14.4	14.9	14.1		
(40) Clear-Sky Solar Irradiance	taub		0.302	0.306	0.306	0.361	0.379	0.392	0.392	0.396	0.360	0.318	0.287	0.278		
	taud		2.225	2.274	2.433	2.332	2.347	2.350	2.343	2.330	2.419	2.516	2.504	2.377		
	Ebn at Noon		243	272	293	283	281	277	275	269	270	268	257	245		
	Edh at Noon		26	30	30	37	38	38	38	37	31	24	20	20		
(44) All-Sky Solar Radiation	RadAvg		526	891	1227	1555	1745	1938	2035	1707	1275	797	491	372		
	RadStd		56	95	126	130	116	130	83	140	84	114	56	42		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (27 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page